



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

20.0 - Application of AI, ML and DL in PETRONAS in supporting Sustainability

July 2025

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Course Content

20. Applications of AI, ML and DL in PETRONAS (in supporting Sustainability)

5

VROC

PREDICTOR (AIC-3261): Prediction using ML

CV CARE – Prediction of Early Degradation of Control Valve

Learning Outcome

At the end of the session, you will be able to:

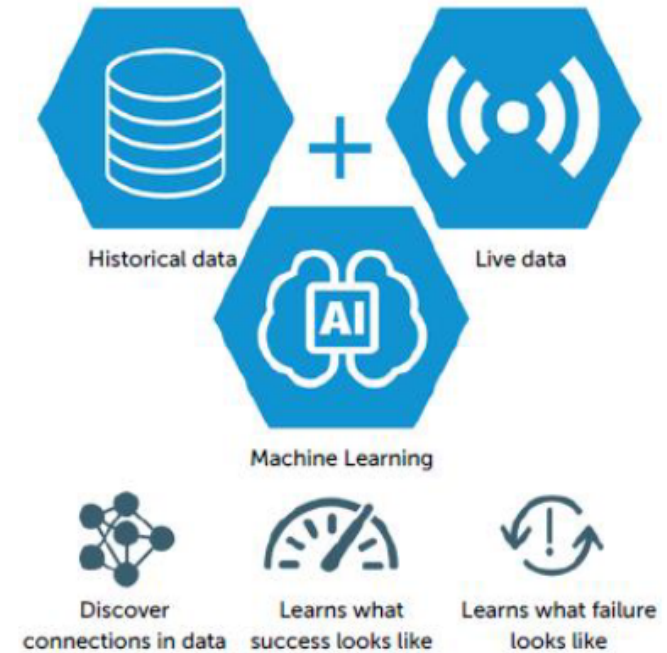
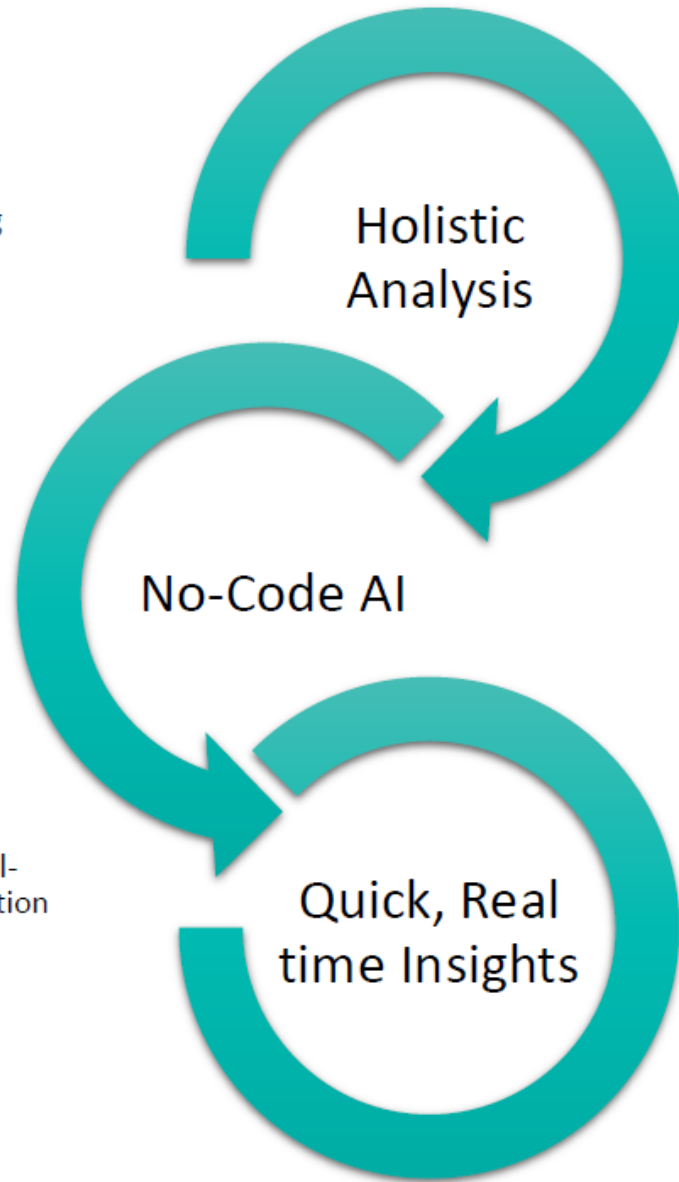
- Explain the basic fundamentals of data statistics and analytics.
- Describe of common terminology on AI and its subsects ML and DL.
- Apply data analytics, AI and their subsets usage in various ICM applications.
- Describe the characteristics of computing requirement most suitable for AI.

Applications of AI, ML and DL in PETRONAS (in supporting Sustainability)

- VROC
- PREDICTOR (AIC-3261): Prediction using ML
- CV CARE – Prediction of Early Degradation of Control Valve

What is AI Machine Learning TTF-VROC?

- TTF-VROC offers holistic analysis by modeling specific equipment and using broad data fields to identify interconnected problems and contributing factors.
- Using the platform's, no-code AI, engineers and subject matter experts can quickly create models that learn from data, such as forecasting, time-to-failure, root cause analysis, and deviation.
- Continuous predictions offer quick, real-time information that enable prioritization of actions, quick answers to business demands for instant gains.



- ✓ Artificial intelligence mimics human problem solving and decision-making capabilities to achieve intuitive outcomes.
- ✓ To build a statistical or mathematical model for AI, large volumes of data are needed to handle the dynamic behaviours of an industrial process.
- ✓ Statistical or physics-based models alone provide a post-mortem fitted result, but AI can quickly solve **multi-dimensional issues** to address **dynamic process behavior**.

Case for change

Assets is using 1st principles and tacit knowledge as their common analytics approaches, which is difficult to pin-point the root cause **due to complexity of the inter-related processes that requires many data points to be analyzed.**

This is seen in the **maintenance** and **process operation interventions** failure analysis for both preventive and reactive measures.

PAIN POINTS

Unable to predict Time to Failure



- Difficulties to identify root cause for **Gas Compressor Modules frequent trips** in 2017 leading to **UPD and value loss** of 325 bbl/hr whenever trip occurs.
- **Reactive maintenance** of turbomachineries post trip resulting in **premature aging** of compressor bundles.
- Lack of monitoring and analytic tool in process operation to **cover the balance of plant.**



Data Driven and Predictive Time to Failure



CASE FOR CHANGE

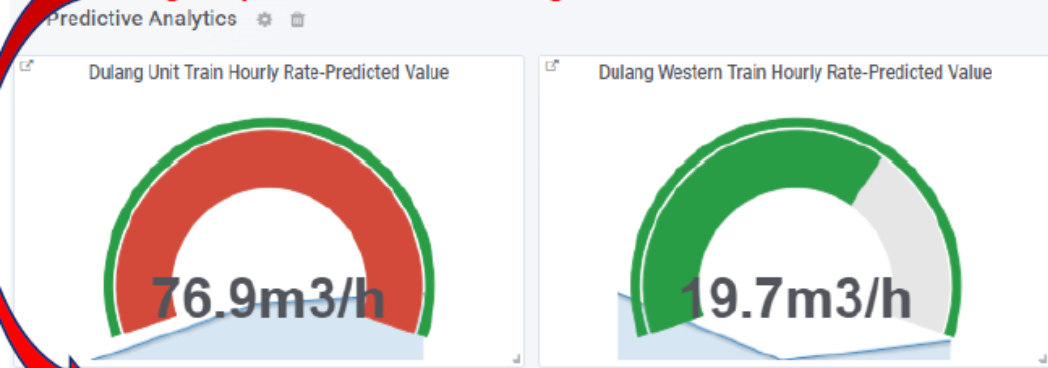
- Data driven analytics utilising AI and Big Data to identify causal factors for various scenarios within a short time.
- Predictive Time to Failure with 7 days lookahead for the assets to avoid Unplanned Production Deferment (UPD)
- Provide adequate time for assets to plan the maintenance activities especially for platforms that adopt remote operation philosophy

Success Story No.1: Dulang High Liquid Production Flow to FSO

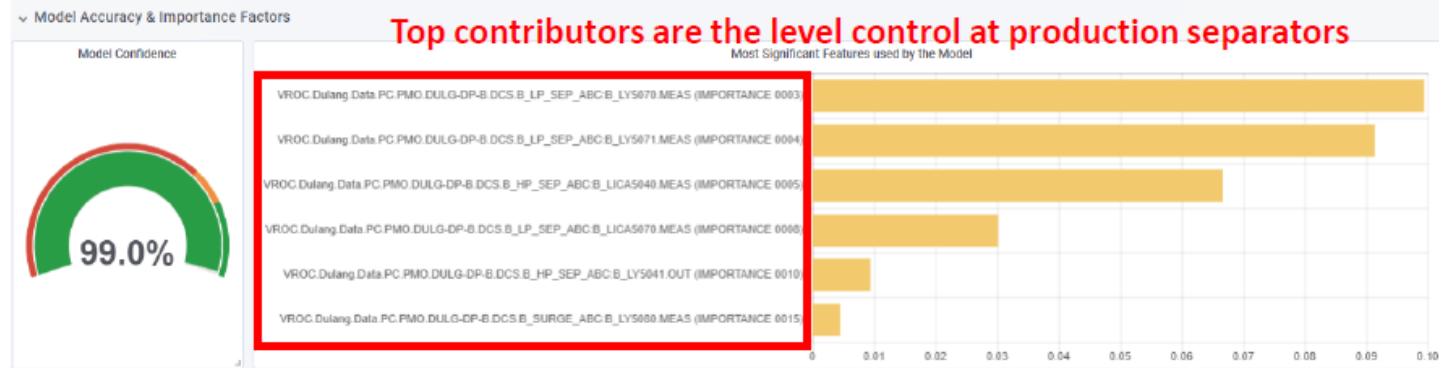
Scenario: Observed sudden spike of hourly crude production figure to FSO in VROC Predictive Analytics Model on 27/2/2023. Suspected produced water has carried over to FSO due to emulsion & erratic level control at production separators.

Findings: From the VROC Model Prediction & Important Factors Trend, SME has observed that the level control in production separators were fluctuating badly. Thus, leading to produced water carried over to FSO and high hourly crude production figure. Operation has performed level control fine tuning immediately to rectify the issue and the crude hourly production figure has back to normal.

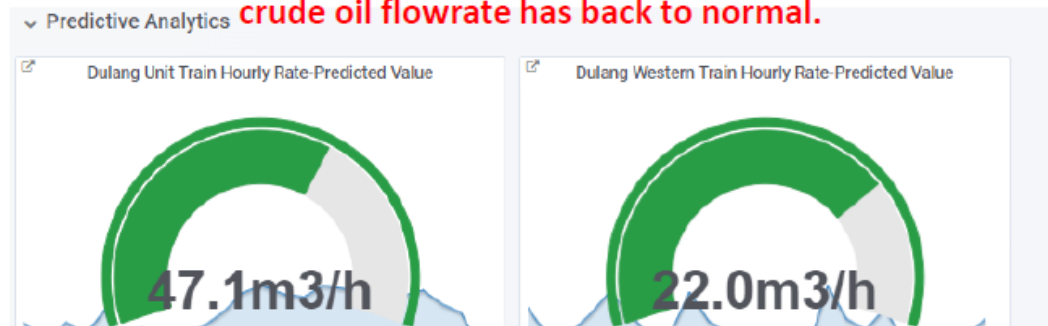
High liquid flowrate sending to FSO on 27/2/2023



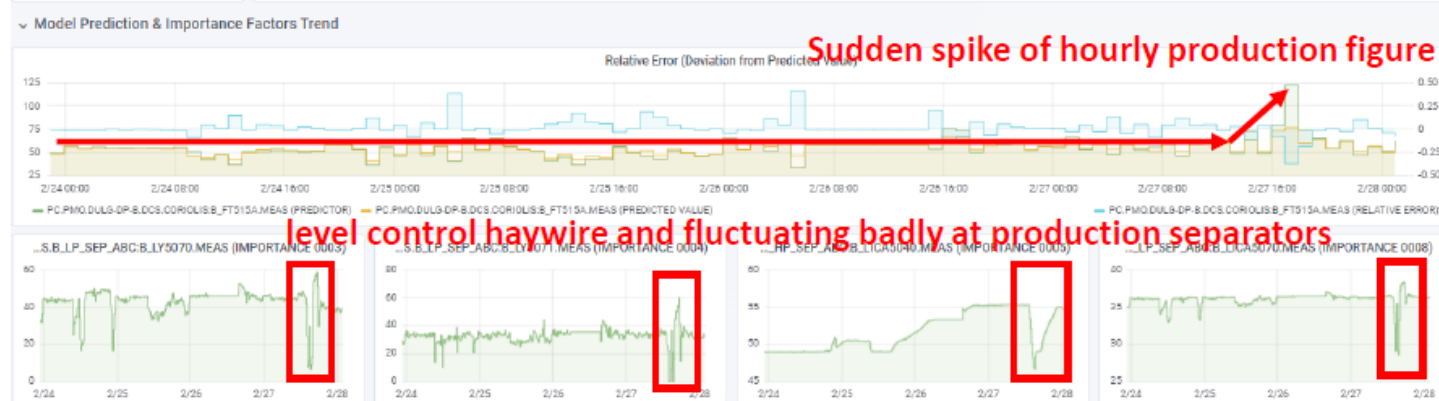
Top contributors are the level control at production separators



After rectification & troubleshooting at site, the crude oil flowrate has back to normal.



Sudden spike of hourly production figure



Challenges of VROC – TTF implementation

1. PI Data Availability and Reliability

- TTF – VROC high dependence on Data Availability and Reliability resulting in incomplete TTF prediction models, i.e RESAK

2 Informal Team Organisation Support Structure

- Lack of a strategic adoption plan to ensure the maximum utilisation
- Current team is based on asset to asset but need a formal support system from Centre, to emulate PROTEAN support structure

3. TTF – VROC user interface and input data tagging system

- Current user interface lack aesthetic features which potentially deters the adoption from the end-users
- Input data are directly taken from PI tag number without descriptors which necessitate end user to be either be familiar with individual tagging or go through another route, i.e open up PI system to check for details.

4. Alert notification (New feature)

- TTF model sending alert notification with a time lag, e.g Dulang GCM trip alert issued 30 minutes after the trip occurred.

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PREDICTOR : Analyser Anomaly Detection by PCMTBE

Business Pain-point

Analysers are very sensitive equipment and sometimes will have inaccuracies or failures during service. A gas chromatograph is a critical analyser for Petrochemical plant. Failure or downtime will result in possible product quality reduction.



Gas
Chromatograph
(GC) Analyser

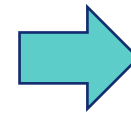
An equipment similar to a small laboratory which measures the components in a product. It is used to determine the quality of components in the sample.



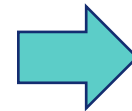
Azure Machine Learning

How Might We...

Predict online gas chromatograph readings using machine learning so that we can provide a secondary measurement in the event of an analyser failure as well as using the predicted values to anticipate measurement error thus avoiding product off specification or quality give away.



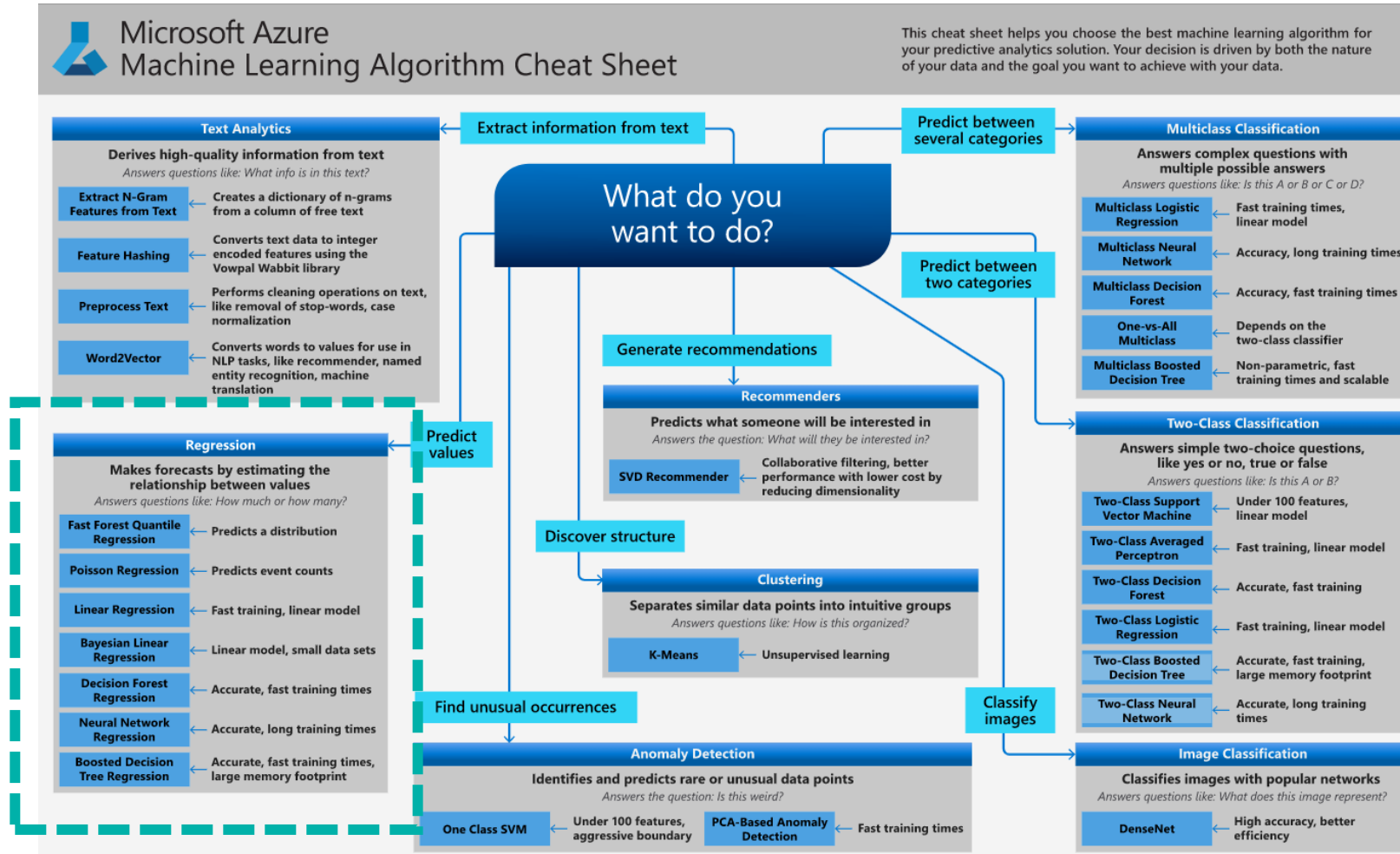
Secondary
Measurement



Avoidance
Measurement
Error

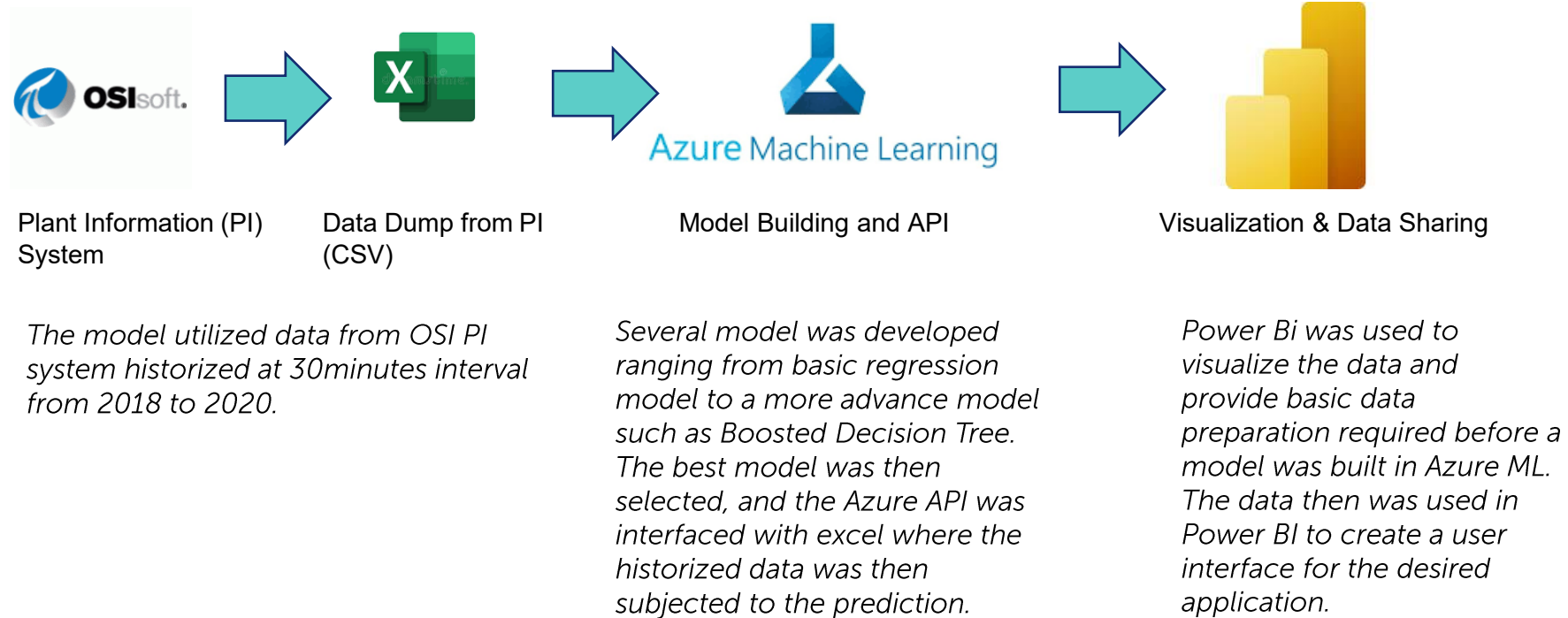
What is machine learning?

Programs that learn from data and apply the learning for an informed decision



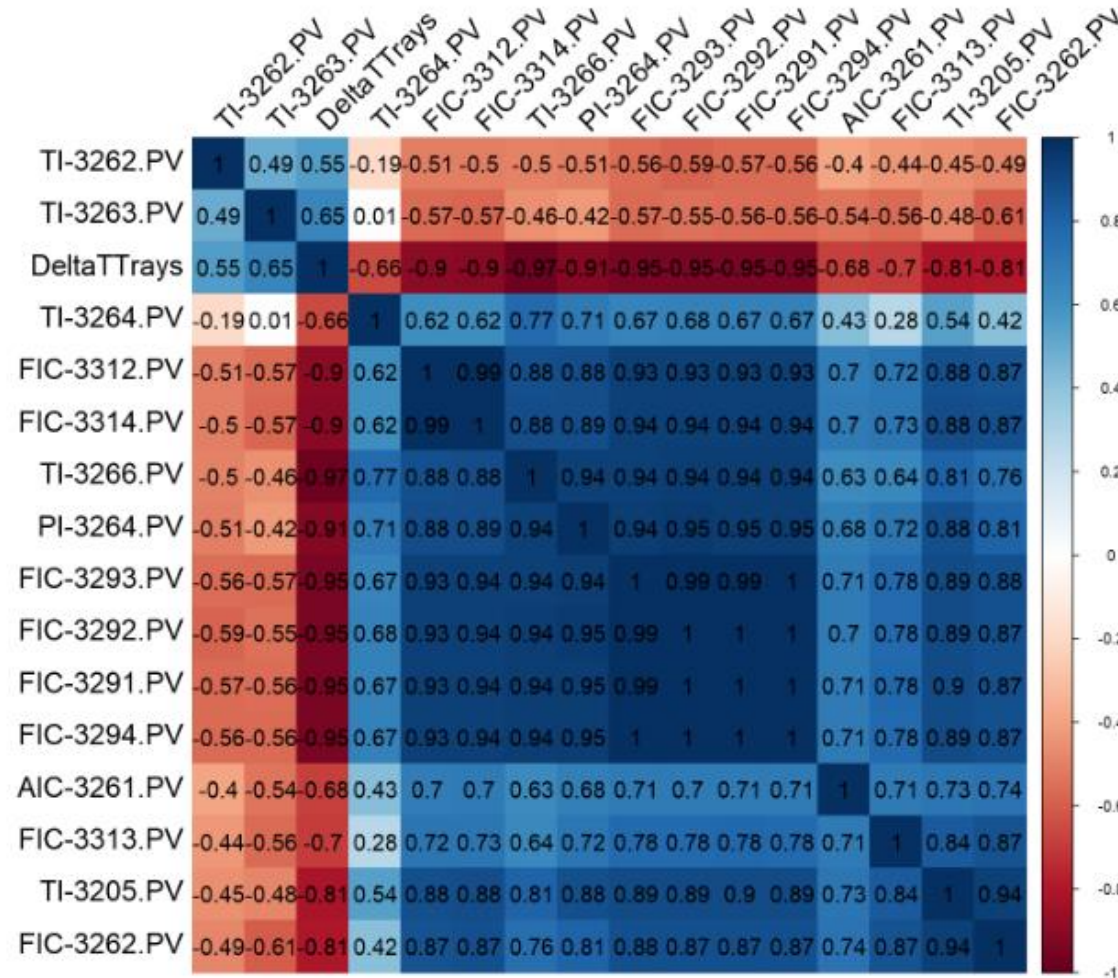
The Approach

Visualizing data using Microsoft Power BI with integration of analytics from Azure Machine Learning



Analytics Mastery: Visualizing Data Relationship

Correlation: Degree of relationship between Variables



Top 5 Correlating Data with AIC3261

1. FIC 3262
2. FIC 3313
3. TI 3205
4. FIC 3291/92/93/94
5. FIC-3312/14

Analytics Mastery: Modelling : Identification of Parameters

Defining parameters based on Correlation, Operational Knowledge and Feature Engineering

	Correlation		Based on Correlation	Based on Operational Knowledge	Feature Engineering & Hyper Parameter Tuning
		AIC-3261.PV			
1	AIC-3261.PV	1			
2	DeltaTTray_TI-3263_3266	0.783146708			
3	TI-3266.PV	-0.775256037			
4	FIC-3313.PV	0.750790966			
5	FIC-3315.PV	0.653474664			
6	FIC-3311.PV	0.638690953			
7	FIC-3262.PV	0.592306256			
8	TI-3263.PV	0.573381047			
9	TI-3205.PV	0.558730757			
10	PDI-3265.PV	0.496747314			
11	FIC-3263.PV	0.496633701			
12	TI-3262.PV	0.474371831			
13	LIC-3261.PV	-0.435467575			
14	LIC-3312.PV	0.27773612			
15	TI-3264.PV	-0.218092163			
16	FIC-3312.PV	0.178426277			
17	FIC-3314.PV	0.177408179			

Boosted Decision Tree Regression	Mean Absolute Error	0.01784	0.015501	0.014876
	Root Mean Squared Error	0.029175	0.02547	0.023694
	Relative Absolute Error	0.143103	0.124346	0.119333
	Relative Squared Error	0.033524	0.025549	0.022112
	Coefficient of Determination	0.966476	0.974451	0.977888

Final parameter selection based on best coefficient of determination

Linear Regression	Mean Absolute Error		0.026562	0.021112
	Root Mean Squared Error		0.03568	0.027773
	Relative Absolute Error		0.213071	0.169352
	Relative Squared Error		0.05014	0.030378
	Coefficient of Determination		0.94986	0.969622

Bayesian Linear Regression	Mean Absolute Error		0.027524	0.023257
	Root Mean Squared Error		0.037276	0.03102
	Relative Absolute Error		0.220784	0.186561
	Relative Squared Error		0.054727	0.037899
	Coefficient of Determination		0.945273	0.962101

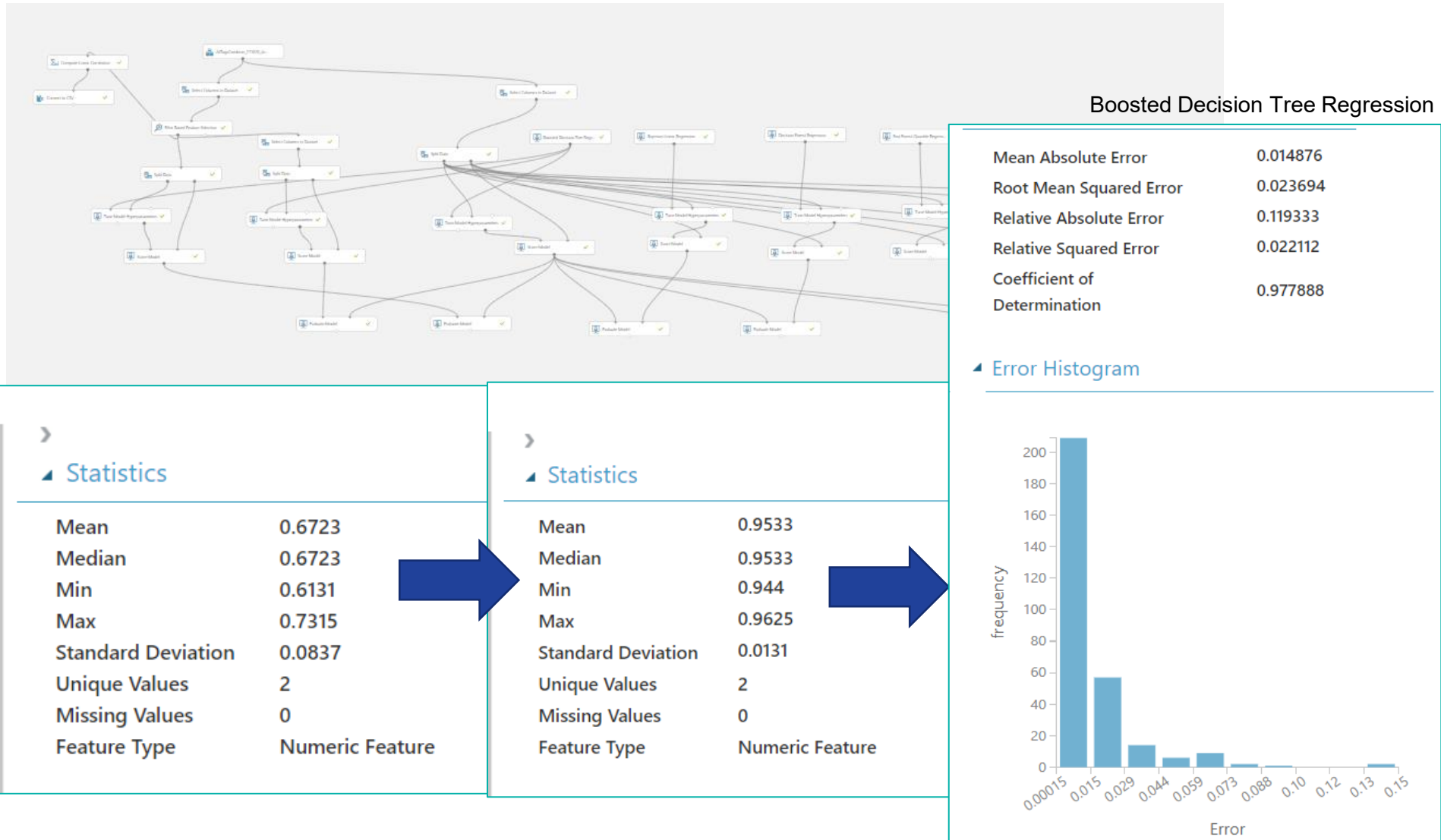
Decision Forest Regression	Mean Absolute Error		0.018275	0.013329
	Root Mean Squared Error		0.037762	0.023736
	Relative Absolute Error		0.148723	0.108477
	Relative Squared Error		0.059525	0.023518
	Coefficient of Determination		0.940475	0.976482

Neural Network Regression	Mean Absolute Error		0.033855	0.030547
	Root Mean Squared Error		0.044226	0.040244
	Relative Absolute Error		0.271574	0.245033
	Relative Squared Error		0.077034	0.063789
	Coefficient of Determination		0.922966	0.936211

Analytics Mastery:

Selection and fine-tuning model based on “Feature Engineering”

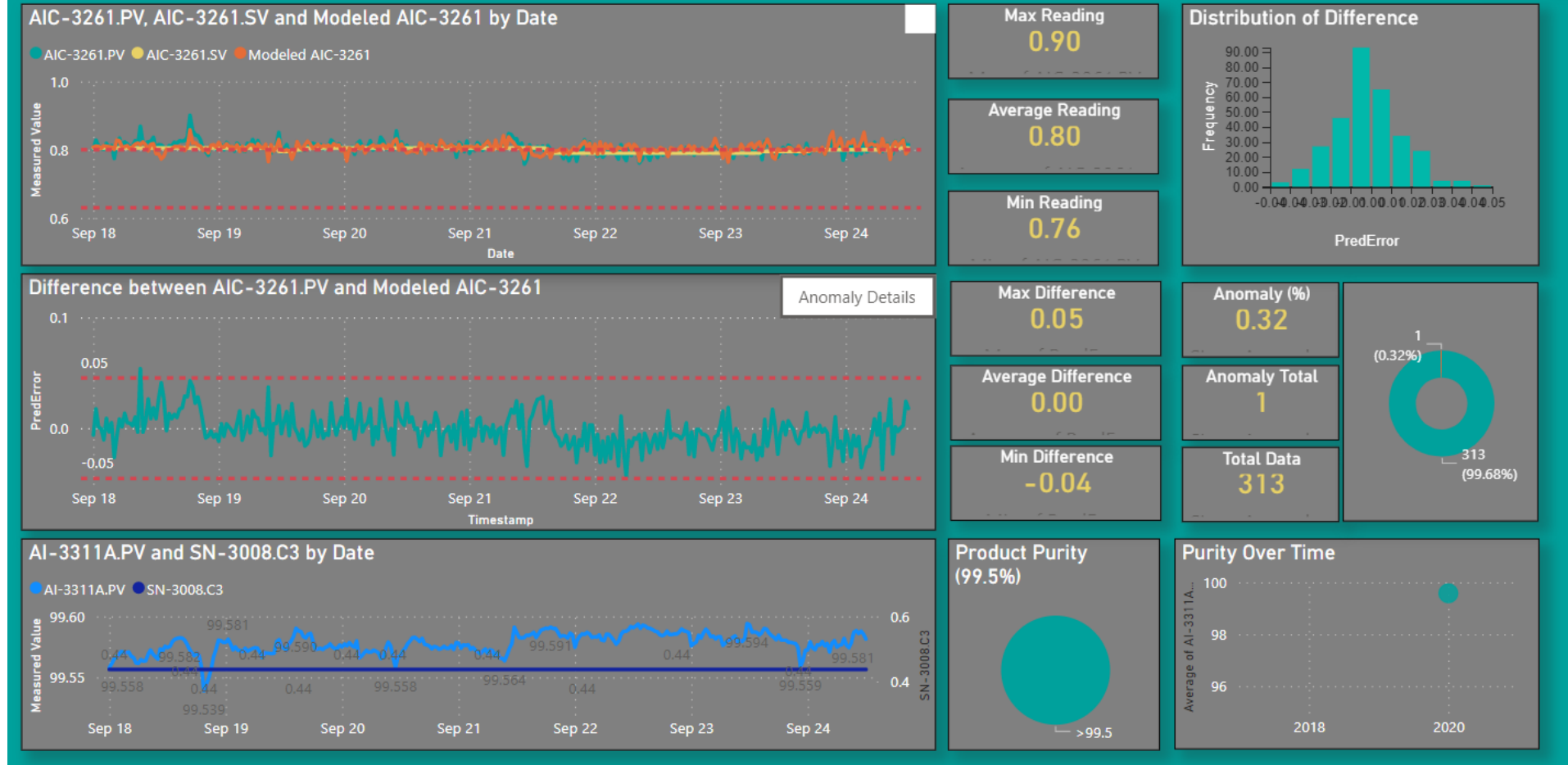
Result from 0.7315 to 0.9778 on Coefficient of Determination



Viability: Model Result & Visualization on Power BI

Model created using Azure ML and represented using Power BI, provide data backed insight.

PREDICTOR : Model Prediction vs AIC-3261 Reading

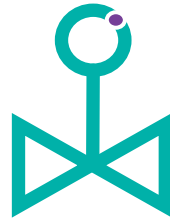


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CV-CARE predicts control valve degradation issues

IRIS Modules



CV-CARE

Control Valve Analytics

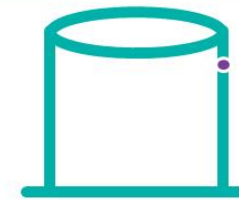
Analytics to predict early degradation issues for control valve



VALVE TRACKER

On-off Valve Analytics

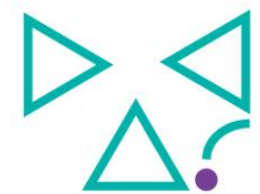
Automatic capture and reporting of IPF Valve Proof Test and monitoring valve performance



IRIS ELFA

Early Level Failure Analysis for ATG

Predicts and alerts electronic failure of Automatic Tank Gauge up to 4 months in advance



IRIS iSOVMS

Intelligent Solenoid Operated Valve Management System

5G-enabled real-time predictive digital solution alerting coil degradation of Solenoid Operated Valve



i-SENSE

Sensor Drift Analytics

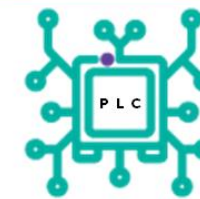
Analytics for detecting sensor drift between monitoring, control and trip transmitters



I-ID

Instrument Inventory Database

Live instrument obsolescence data for efficient instrument life cycle and budget management

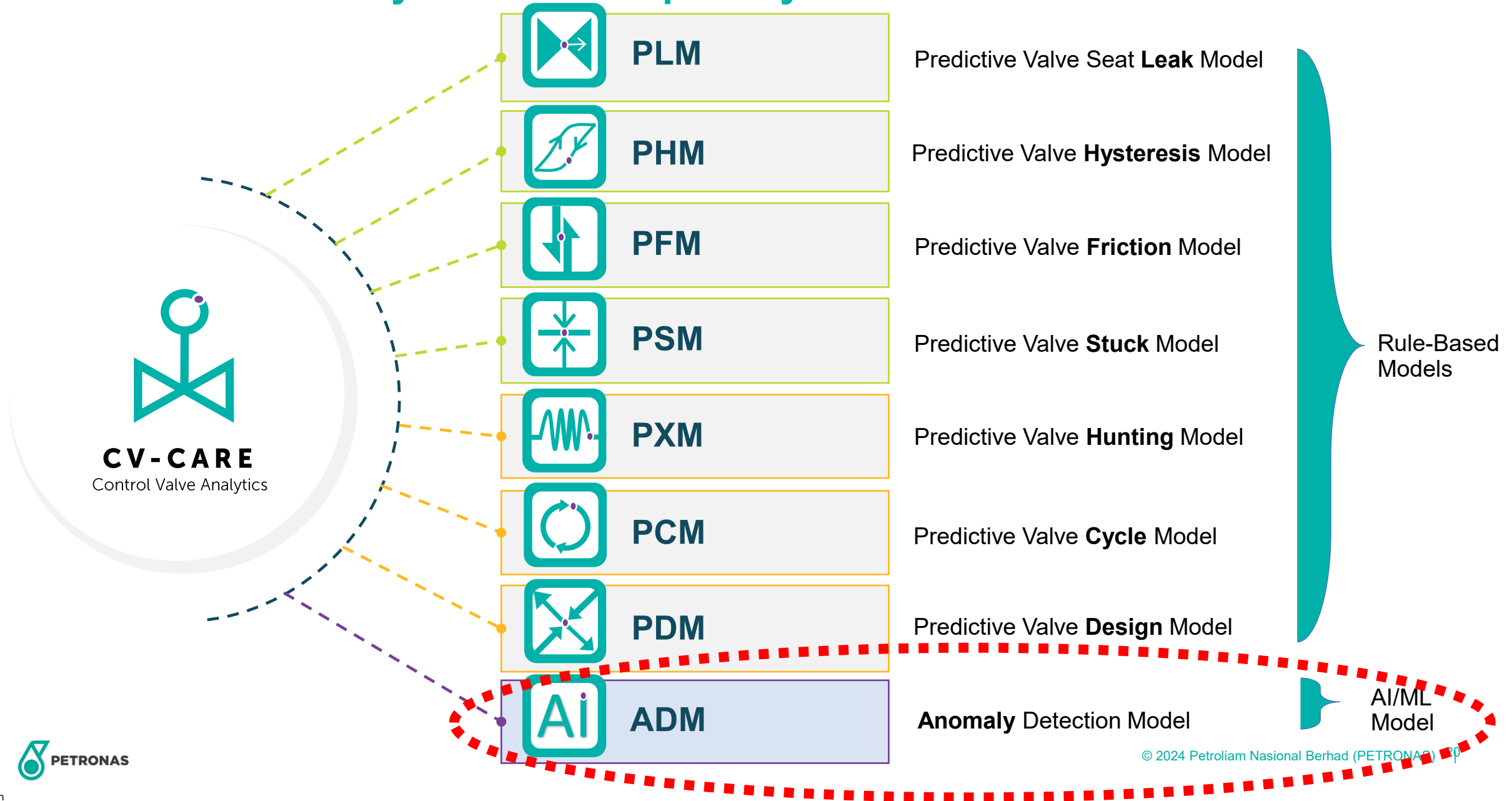


SCREEN

Sub-system Controller Surveillance

Centralized visibility of PLC diagnostics for early intervention of faults

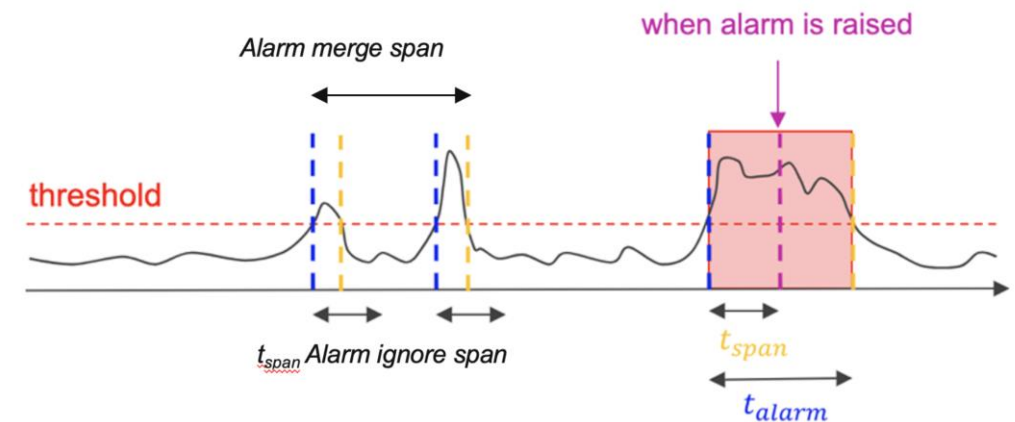
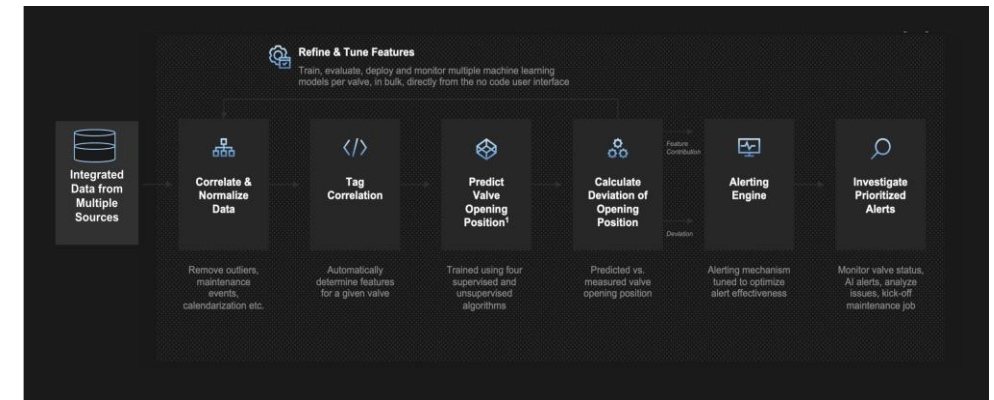
CV-CARE's Anomaly Detection capability uses AI/ML



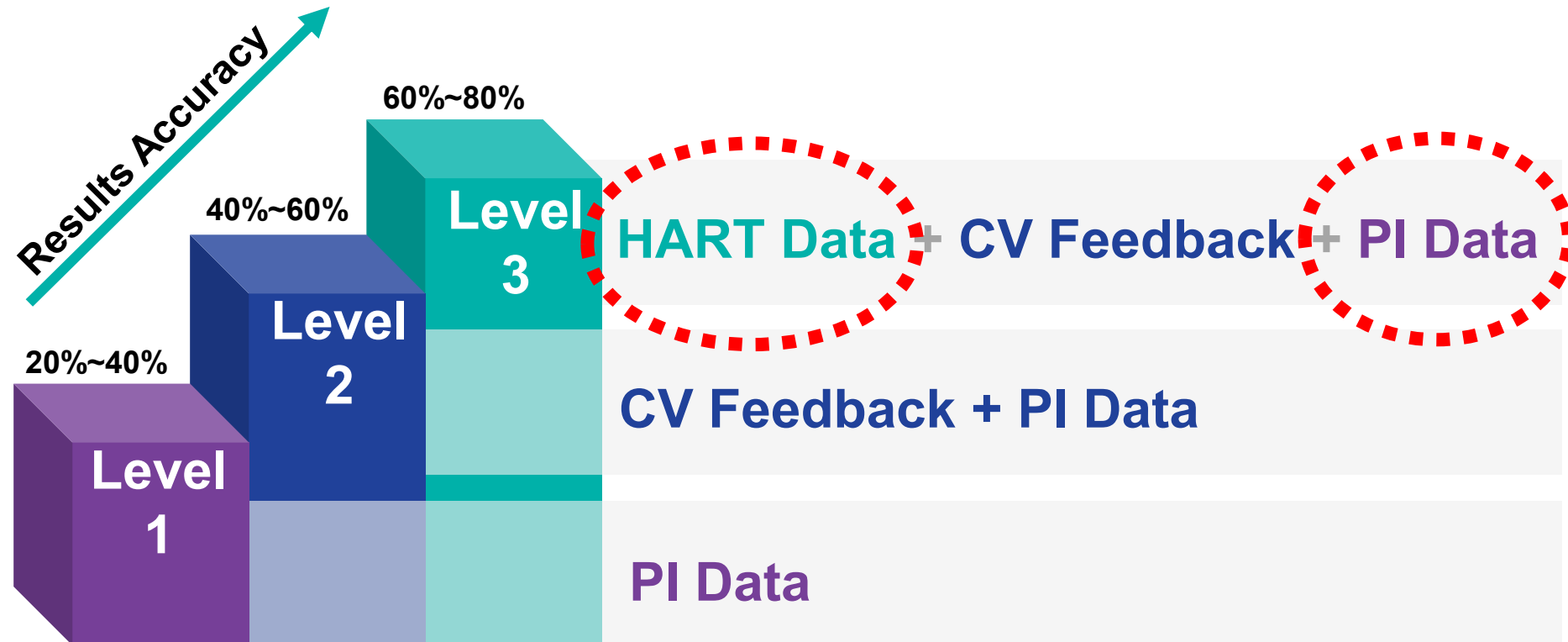
CV-CARE's Anomaly Detection uses Regression



Predict CV position using semi supervised approach with regression modeling to generate alert based on comparison against CV position based on predetermined criteria.



Inputs required by CV-CARE's Anomaly Detection Model are HART Data & PI Data



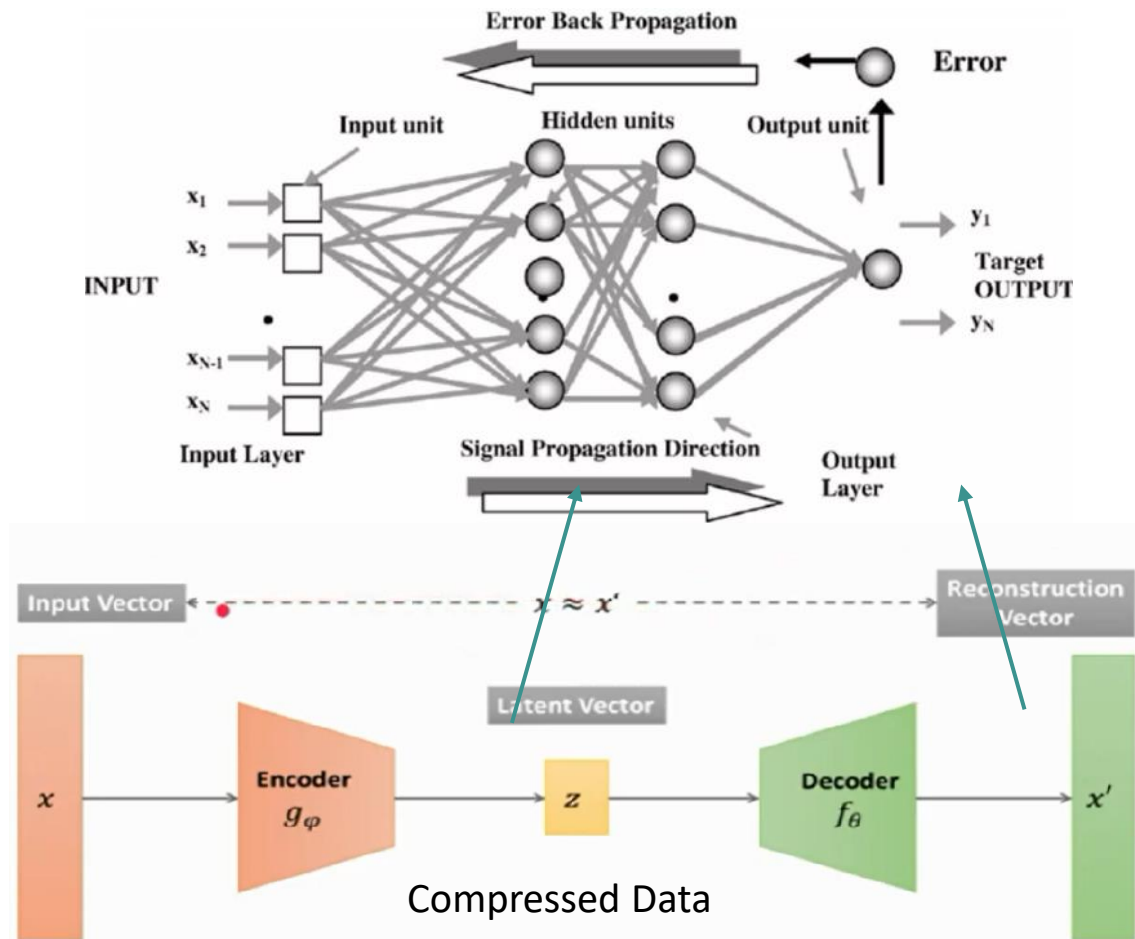
ADM was able to predict Valve Stuck 15 days ahead

ADM & PSM Models' Interaction



Autoencoder Concept

Autoencoder is one of the powerful algorithms under deep learning umbrella to detect anomalies. It is a type of unsupervised deep learning model that use hidden layers to decompose and then recreate their input.



The process for autoencoding in anomaly detection can be summarized as:

- Feed historical normal data through encoder network i.e. any outliers from the historical data need to be filtered out
- Generate the lower dimension embedding (backend process)
- Feed embedding through decoder network (backend process)
- Generate reconstructed version of the original data
- Compare the result of the generated vs the original image

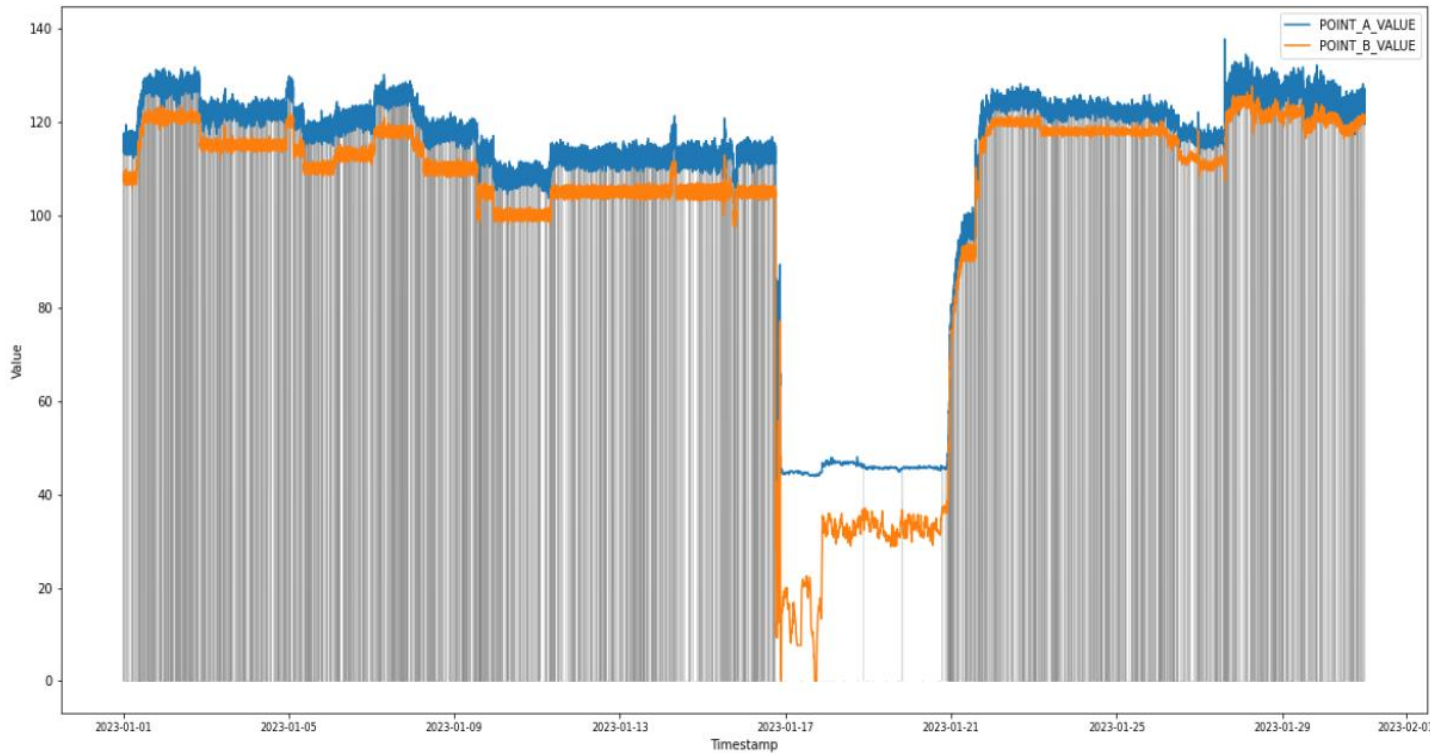
Since, the model has been fed with normal data reading, the reconstructions definitely are normal data predictions. Any discrepancy between actual and reconstruction reading will be calculated as a risk. Alert will be raised if the risk is more than the set threshold. As of now, the threshold is **0.9**.

Data Masking

Criteria:

1. High fluctuation data is removed from the training datasets
2. Plant Load level used to baseline and detect changes

Data selected for model training



Control Valve Data Set

DCS - 21FC217.DACA.PV
DCS - 21FV217.DACA.MV

Masking findings:

1. Grey highlighted area is the area where the data is masked and selected for training the model

